

Stack Overflow Developer Survey Analysis

Data Analyst Capstone Project

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Executive summary

Key findings from survey, job posting, technology, and compensation analysis

65,437

Survey responses

114

Variables analyzed

185

Countries represented

\$65,000

Median yearly comp.

Current technology usage is led by JavaScript and PostgreSQL.

Future learning interest is strongest for Python and PostgreSQL.

Compensation is highly right-skewed; log-scale and outlier-aware charts provide clearer salary insights.

Dashboard screenshots are the only major missing items before final submission.

Project overview

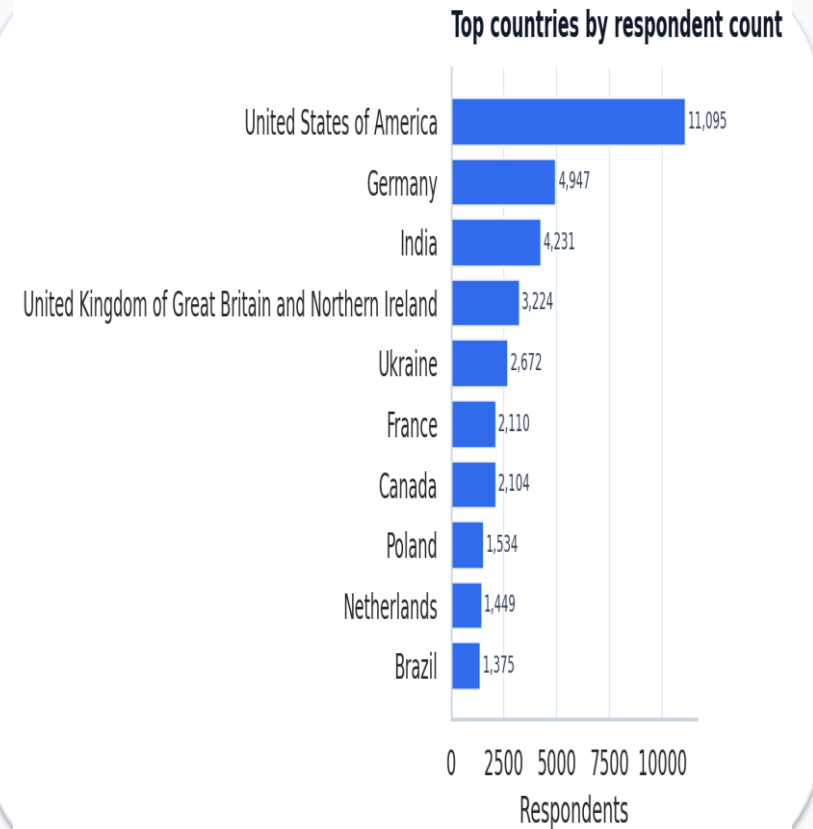
Purpose, audience, and business value

Purpose: identify current technology usage, future learning trends, and compensation patterns among developers.

Audience: learners, hiring teams, and career changers planning skill development and technology focus areas.

Value: connects developer preferences with job posting demand, salary trends, demographics, and remote-work patterns.

Main analysis areas: programming languages, databases, employment, remote work, age, country, compensation, and job satisfaction.



Methodology

Data preparation and analysis approach

Data sources: Stack Overflow survey dataset, job-postings.xlsx, technology_jobs.xlsx, and popular-languages.csv.

Wrangling: handled missing values, converted numeric columns, split semicolon-separated technology responses, and filtered outliers where needed.

Analysis: frequency counts, median salary comparisons, distribution plots, box plots, scatter plots, and trend comparisons.

Visualization: charts were selected to match the IBM template while keeping the deck clean and readable.

Job posting demand

Locations with the highest number of job postings



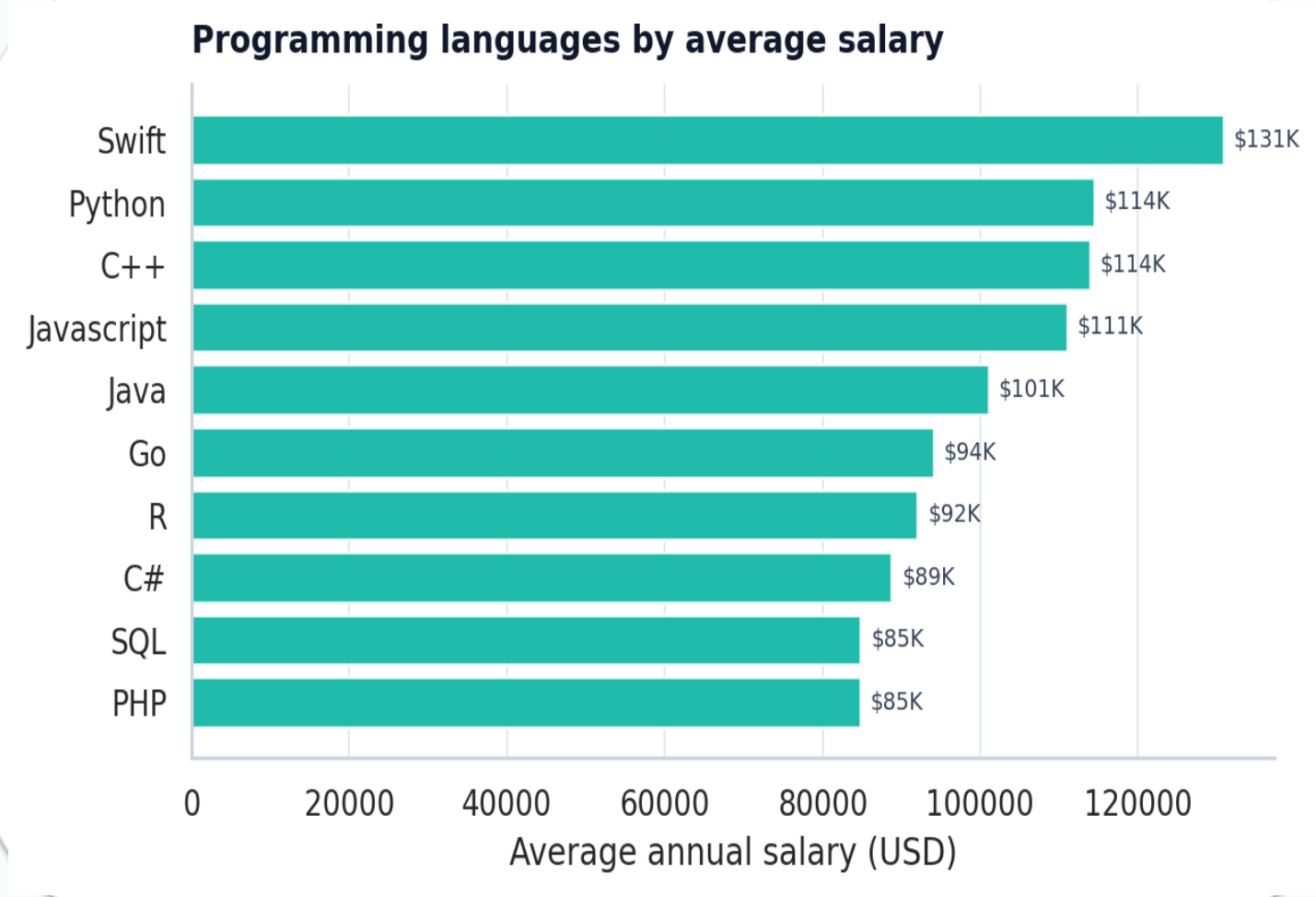
This slide satisfies the template requirement for job-postings.xlsx.

Job postings are ordered in descending order to show the strongest hiring locations.

Use this as labor-market context beside developer survey trends.

Popular languages by salary

Average annual salary from the scraped programming language dataset



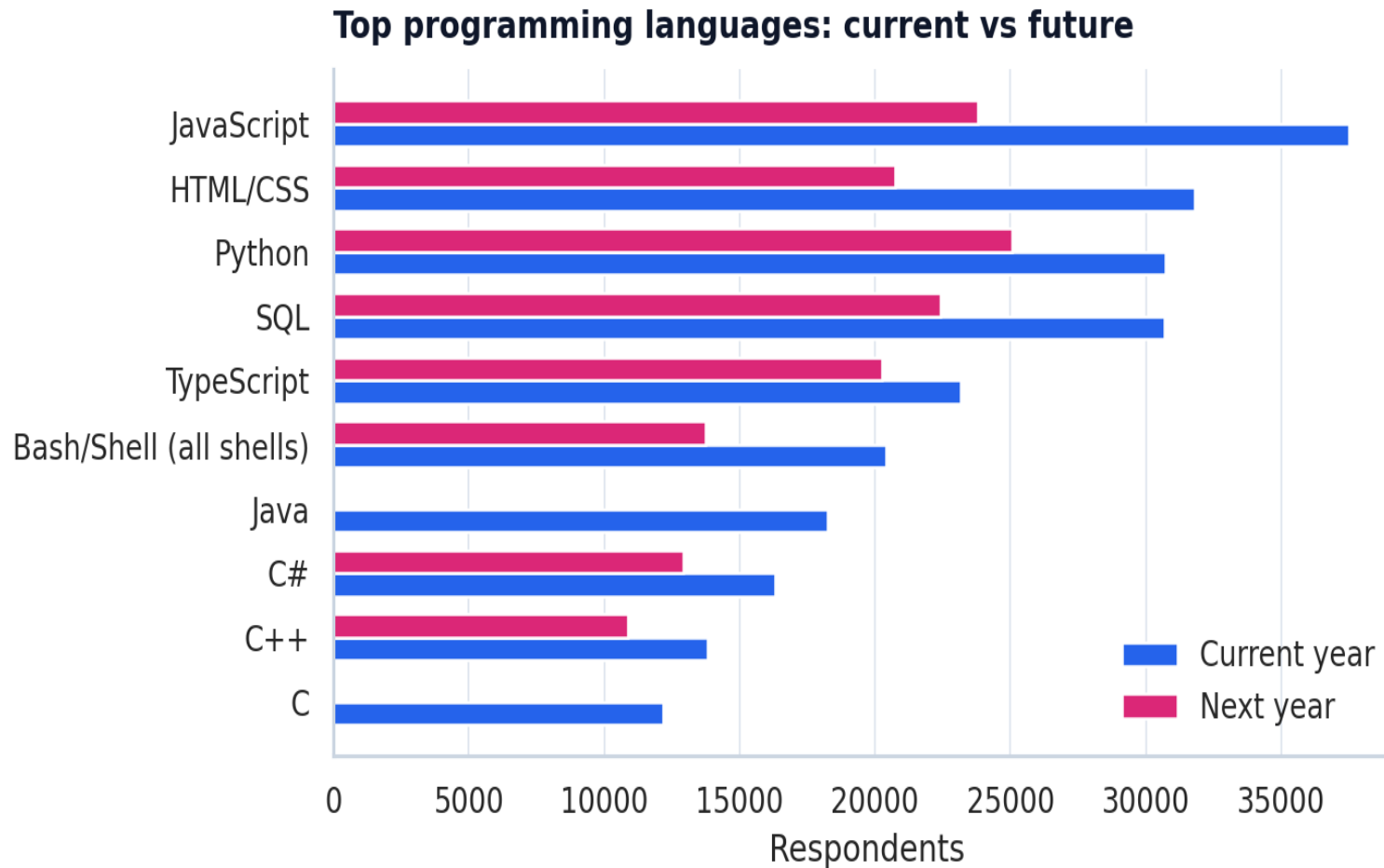
This slide satisfies the template requirement for popular-languages.csv.

Languages are sorted by average annual salary.

Salary context helps connect technology skills with career opportunity.

Programming language trends

Current usage compared with future interest



Current top language: JavaScript.

Most desired next-year language: Python.

High overlap between current and future technologies indicates durable skill value.

Programming language findings & implications

Finding 1: Current usage is led by JavaScript, HTML/CSS, Python.

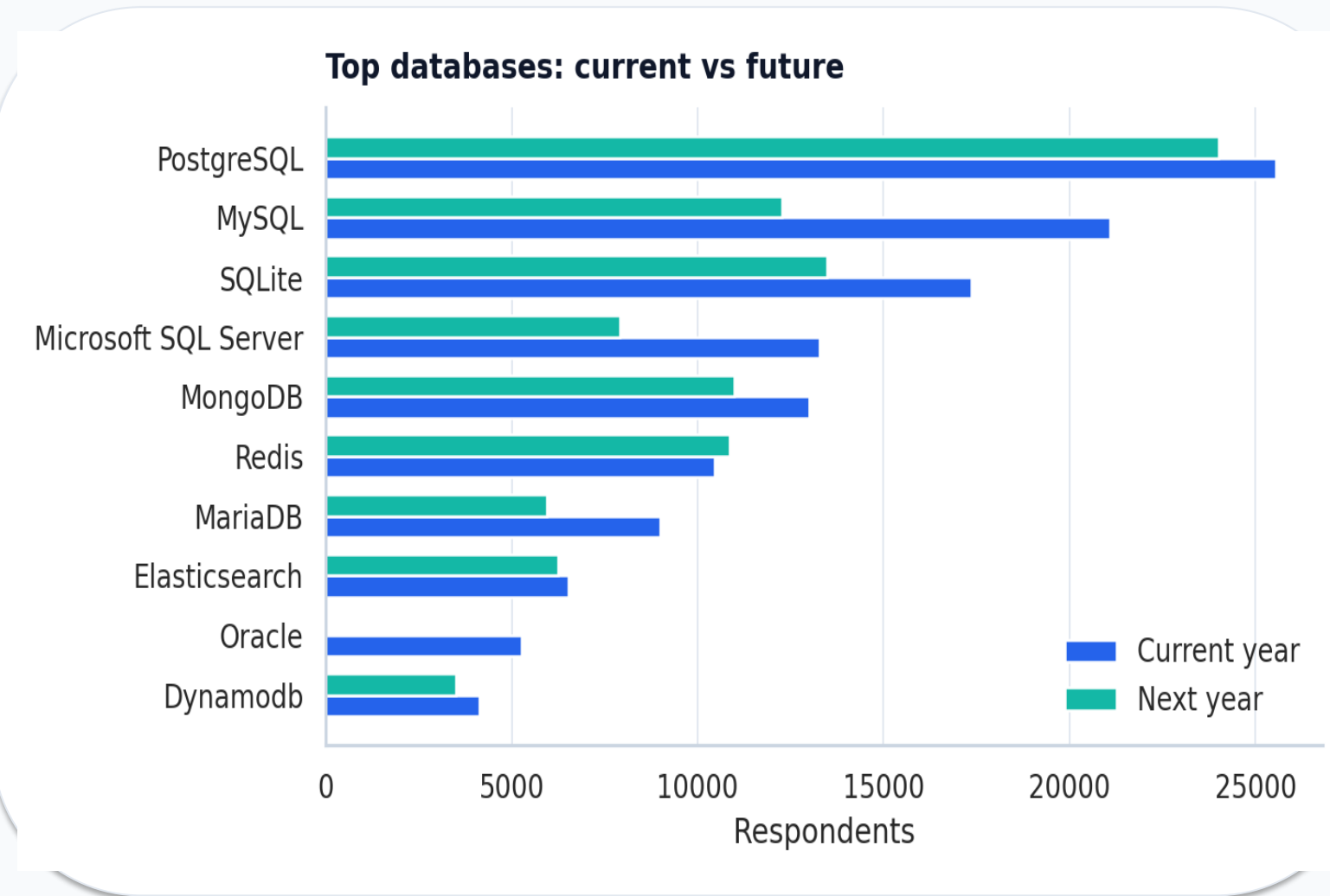
Finding 2: Future interest is led by Python, JavaScript, SQL.

Finding 3: Mature languages remain dominant, while future-interest patterns highlight where developers want to grow.

Implication: Learners should prioritize foundational languages with broad demand, then add high-interest languages to differentiate themselves.

Database trends

Current database usage compared with future interest



Current top database: PostgreSQL.

Most desired next-year database: PostgreSQL.

Database fundamentals remain highly relevant for analytics and software roles.

Database findings & implications

Finding 1: Current database usage is led by PostgreSQL, MySQL, SQLite.

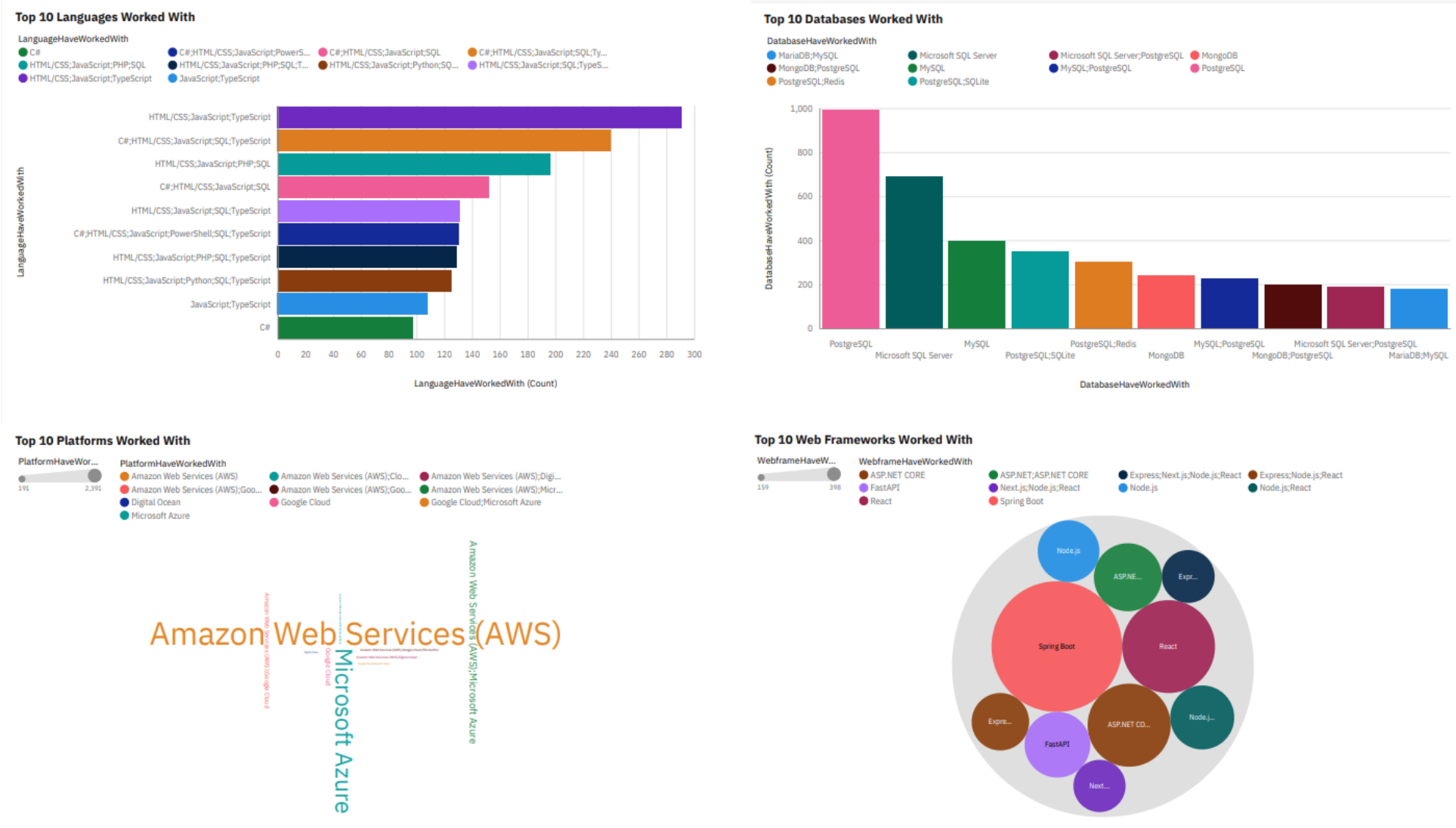
Finding 2: Future database interest is led by PostgreSQL, SQLite, MySQL.

Finding 3: Strong overlap between current and desired databases suggests continued demand for established SQL-based tools.

Implication: SQL and database skills should remain a central part of a data analyst learning path.

Dashboard tab 1: Current technology usage

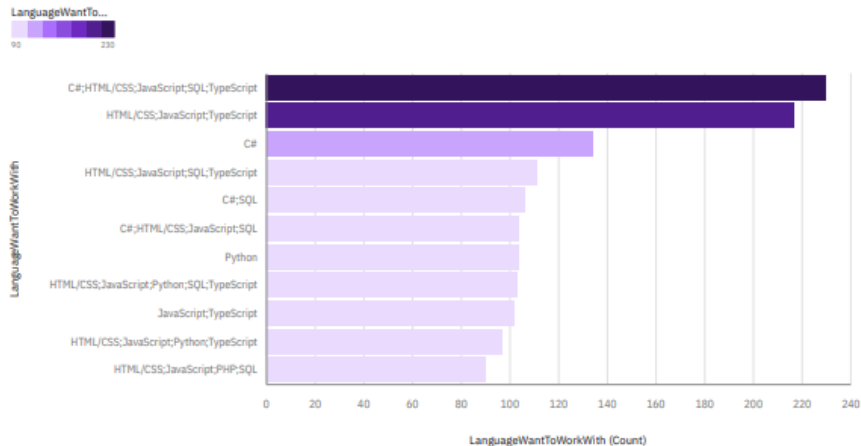
IBM Cognos dashboard screenshot: languages, databases, platforms, and web frameworks used



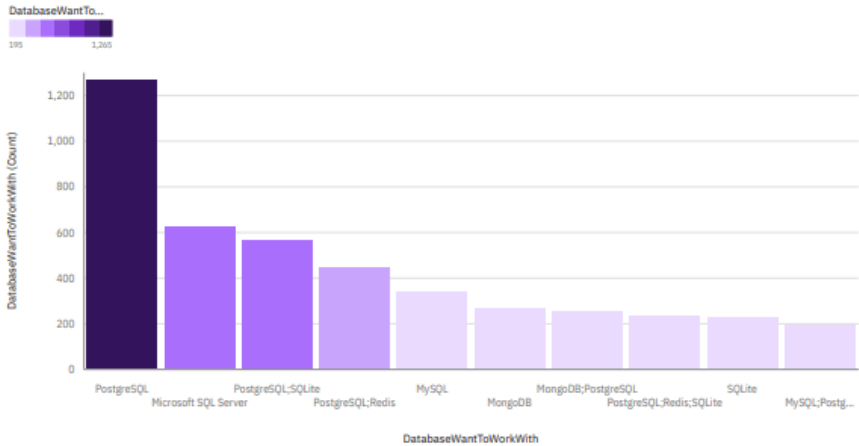
Dashboard tab 2: Future technology trends

IBM Cognos dashboard screenshot: desired languages, databases, platforms, and web frameworks

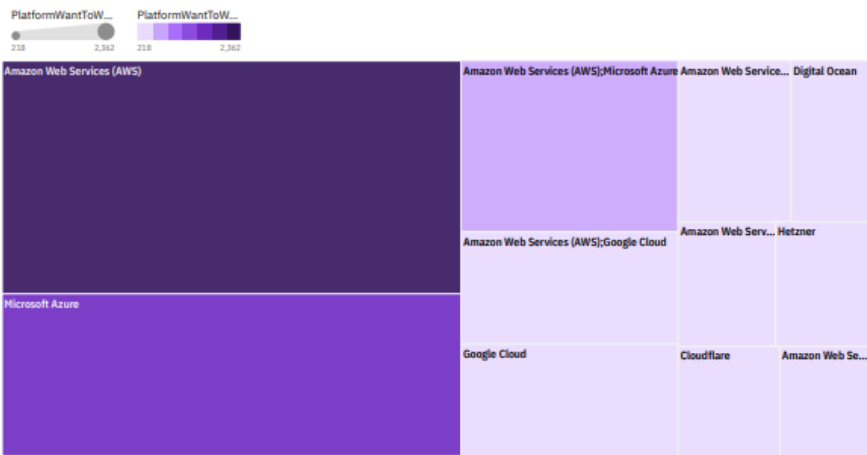
Top 10 Languages Desired Next Year



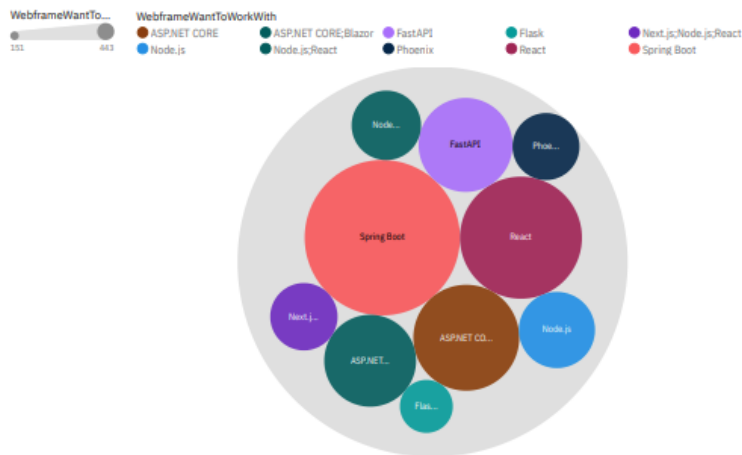
Top 10 Databases Desired Next Year



Top 10 Platforms Desired Next Year

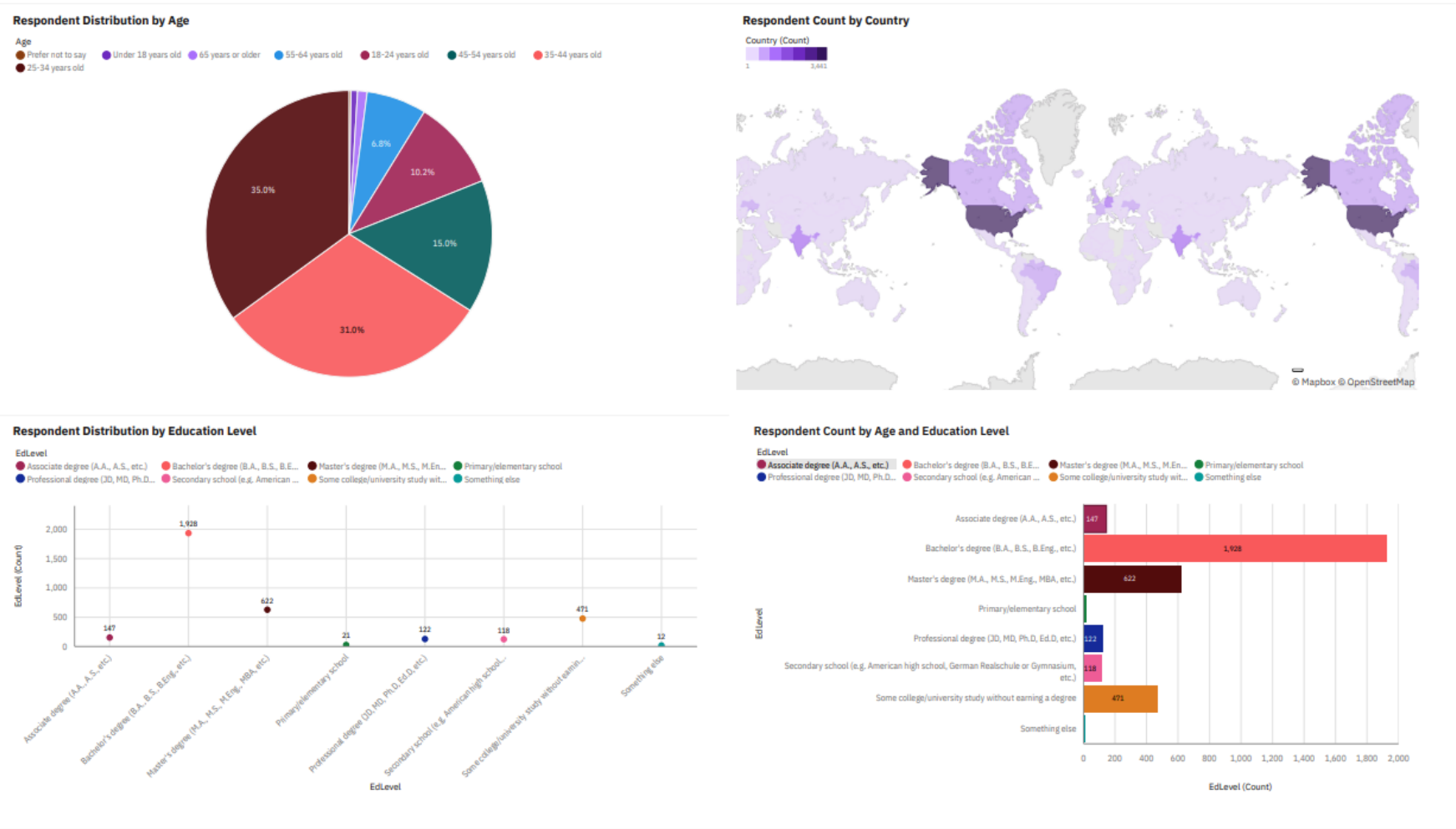


Top 10 Web Frameworks Desired Next Year



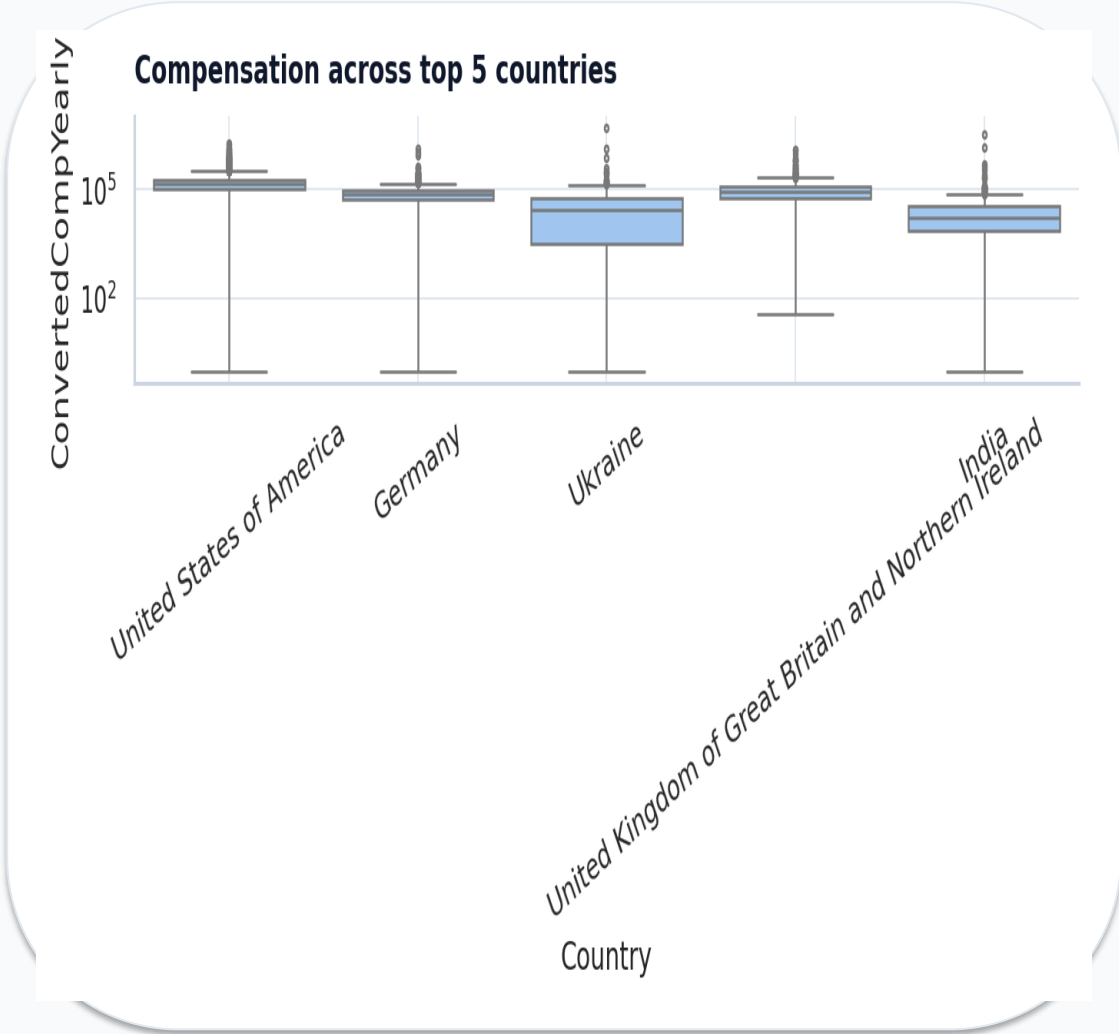
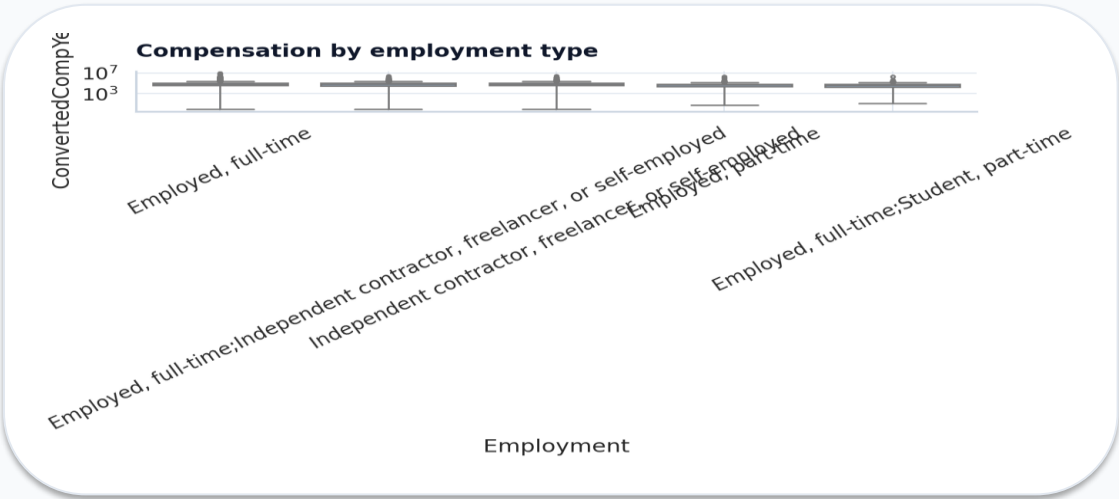
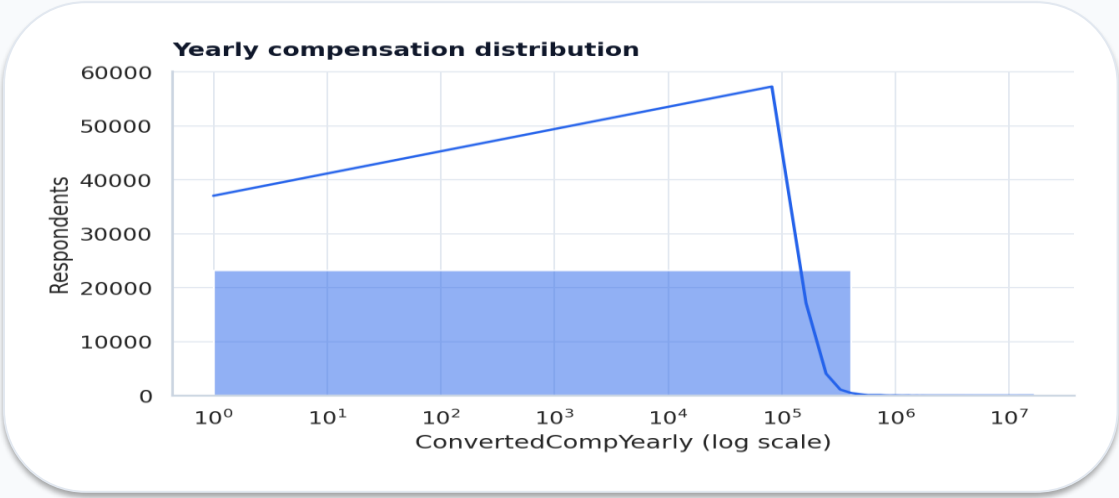
Dashboard tab 3: Demographics

IBM Cognos dashboard screenshot: age, country, education level, and age by education



Compensation insights

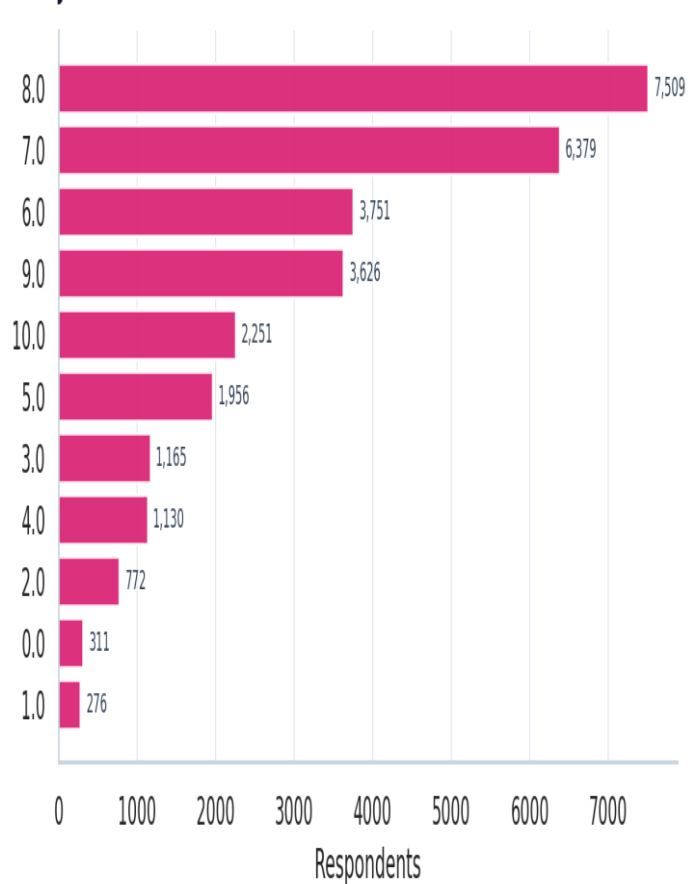
Distribution and variation in yearly compensation



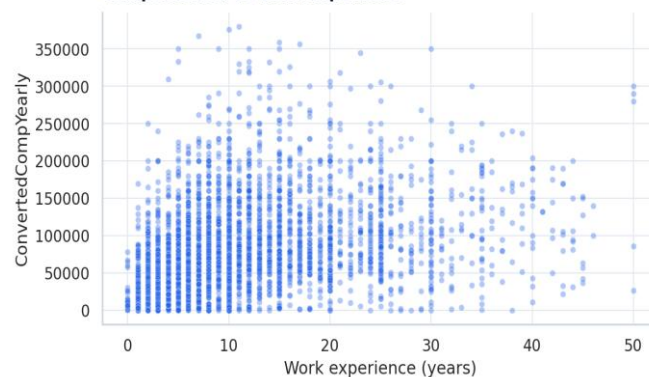
Job satisfaction and experience

Relationships between satisfaction, compensation, and work experience

Job satisfaction distribution



Compensation vs work experience

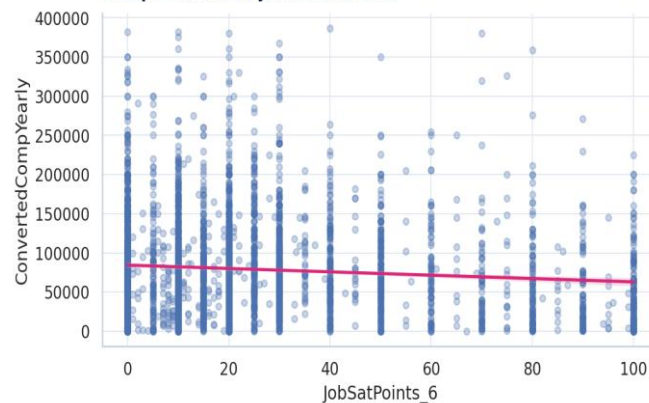


Job satisfaction varies across respondents and should be interpreted with context.

Compensation and experience show visible patterns but also high variability.

Salary outliers can distort trends, so cleaned and log-scale views are recommended.

Compensation vs job satisfaction



Discussion

Overall interpretation of the analysis

Developer technology demand is concentrated around widely adopted programming languages and databases.

Future-interest trends show that developers are continuing to invest in core data, web, and backend technologies.

Compensation analysis provides useful directional insight, but salary data is strongly right-skewed and contains outliers.

Demographic and remote-work trends add context for understanding how the developer population is distributed.

Overall findings & recommendations

Finding: Core programming and database technologies dominate both current usage and future interest.

Finding: Job postings and salary data add external market context to survey-based trends.

Finding: Compensation varies meaningfully by country, role, employment type, and experience.

Recommendation: For career growth, prioritize SQL, Python, database fundamentals, and a small number of high-demand complementary tools.

Recommendation: Use dashboard screenshots to tell the final story visually and satisfy the project rubric.

Conclusion

The analysis identifies key developer skills, future learning priorities, and compensation patterns.

The strongest project story is the link between current skills, desired skills, labor-market demand, and salary context.

The deck is now redesigned in a clean, contemporary style without the large watermark or repeated template color blocks.

Final step: insert the three required dashboard screenshots before submission.

Appendix: Additional analysis

